Fiscal Year:	FY 2012	**Task Last Updated:**	FY 12/02/2011
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Project Title:	HZE-induced mammary cancer development processes in murine and 'humanized' models, and their influence on radiation quality functions		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Comments:			
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Contact Monitor:		**Contact Phone:**	
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Flight Assignment:	NOTE: Received extension to 1/31/2013 per NSSC information (Ed., 2/22/2012)		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Ullrich, Robert (University of Texas Medical Branch)		
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Task Description:	The proposed research will use a model of breast cancer development in which mouse or human mammary stem cells are used to grow mammary glands in mouse mammary fat pads. When the stem cells used are irradiated prior to transplant, progression from very early stages of cancer (ductal dysplasias) can be monitored. We will use this model to compare the development of cancer from cells exposed to gamma-rays versus cells exposed to HZE particles that are found in galactic cosmic rays. The histological progression of pre-cancerous lesions will be monitored using markers of cell growth, senescence, apoptosis, proliferation and angiogenesis. This information will also be correlated with global gene expression profiling of non-persistent and persisting dysplasias to gain further insight into the molecular mechanisms driving progress towards cancer after exposure to the different types of radiation. Studying the cancers that are initiated by radiation in both mouse and human mammary cells will allow a direct comparison of cancer processes that may differ between species as well as between radiation types. This is important, as most assessments of human space radiation risk are based on experiments in rodents.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will provide insight into the cancer risks arising from exposure to space radiation. It asks questions about how cancer is started by radiation exposure, how cancer progresses, and how these processes might be the same and different after exposure to radiation encountered in space, and the more common gamma-rays encountered on earth. Since breast cancer is one of the most common cancers associated with radiation exposure, this is an important model system. Understanding the progression of breast cancer after exposure to space radiation will help NASA to make better risk estimates for their long-term exploration missions. A better understanding of the factors that control breast cancer progression in general can also have a large impact on Earth, by suggesting better intervention or treatment strategies for this very common disease.
Task Progress:	The goal of Specific Aim 1 is “to quantitatively compare early tumor initiating events induced by HZE vs. gamma rays in murine mammary epithelial stem cells using the well-established ductal dysplasia model as well as a 3D in vitro model.” Work on this aim initially used the classic ductal dysplasia model to quantify initiated cells as measured by the frequency of persistent dysplasias in cleared mammary fat pads 16 weeks after injection of mammary epithelial cells. The mammary cells were obtained from 12 week old female BALB/c mice 16 weeks after irradiation with Cs-137 gamma rays, SPE 1972 protons, or 1 GeV/n Fe-56 ions. Following injection of cells, the frequency of persistent dysplasias was quantified 16 weeks later. This model was heavily dependent upon the ability to obtain a consistent number of mammary outgrowths from experiment to experiment in order to quantify dysplasias. Unfortunately, there has been a wide variation in the ability to obtain fully developed mammary glands using this model. In some studies more than 90% of the mammary fat pads injected had outgrowths while in others as few as 50% yielded evaluable mammary outgrowths. As a result data were quite variable and thus required us to perform many more experiments to obtain reliable results. The mice from the experiments performed last year also had a high rate of infection, necessitating more repeated experiments. In last year’s report, we mentioned that we were developing and comparing new approaches for quantification of initiated cells and development of lesions. Fortunately one of these, an intraductal transplantation model based on a recent paper by Behbod et al. (Breast Cancer Res, 11(5): R66, 2009), has proved much more efficient than the initial cleared fat pad approach. We have found that intraductal injection of murine mammary epithelial cells provides similar data to that previously obtained using our dysplasia model (Adams et al. Cancer Res 47(16): 4425-4431, 1987) but requires less time, is more reproducible, and provides more tissue for analysis. Furthermore, based on Behbod et al. 2009, this approach should enable direct comparison of radiation effects in murine versus human breast epithelial cells. Our new approach is not dependent upon the development of mammary glands in order to quantify the dysplastic nature of injected cells. The frequency of dysplastic outgrowths scored in the intraductal injection model tracks well with the frequency in the cleared fat pad approach in studies in which greater than 90% of mammary fat pads injected produced mammary outgrowths. Histological characterization of the outgrowths with respect to their dysplastic nature was performed in these proof-of-principle experiments. Approximately 90% of the outgrowths were classified as dysplastic while 10% could be classified as ductal carcinoma in situ. Both classifications support the interpretation that the cells from which the outgrowths were derived are preneoplastic in nature. Another problematic issue has been the number of dysplastic cells that can be obtained for molecular analysis. Using the previous model, dysplastic outgrowths often involved only a few cells within a mammary duct. In the newer model, the number of cells that make up the dysplasia is higher because the outgrowths are confined in a well-defined area. Importantly, the number of persistent dysplastic growths (fully initiated cells) is higher in mice transplanted with mammospheres derived from mice irradiated with Fe particles compared to those irradiated with gamma rays. This supports our proposed hypothesis, that in comparison to low LET gamma-rays, HZE particle exposure would produce more efficient conversion of mammary cells to a pre-neoplastic state, similar to the response previously seen for high LET neutron exposures. The goal of Specific Aim 2 is “to assay both phenotypic progression and molecular profiles using the mammary ductal dysplasia and 3D stem cell models to determine whether HZE irradiation results in qualitatively different processes, compared with gamma-ray irradiation, that point to different mechanisms of carcinogenesis.” For these studies we are using the dysplasia assay to measure both total dysplasias as determined at around 4 weeks following intraductal injection of donor mammospheres and persistent dysplasias as quantified at around 12 weeks after injection as described under Specific Aim 1 above. Studies currently underway include both histological and molecular characterization from the same outgrowths. In the first studies, cells from gamma irradiated mice and controls were injected using the intraductal model and dysplasias were harvested from mice after 12 weeks. Outgrowths will be characterized histologically and cells from dysplastic outgrowths and normal cells from the same mammary fat pad will be characterized at the molecular level using expression arrays. We also have RNA from the primary and secondary mammospheres from which these outgrowths were derived for characterization on gene expression arrays. In initial analyses, we have compared expression of genes considered putative stem cell markers (Klf4, Nanog, Wnt1) and epithelial cell markers (cytokeratin 14 - basal myoepithelial, cytokeratin 18 - luminal) in the primary and secondary mammospheres used for injection. We found that the stem cell markers Klf4 and Wnt1, as well as the basal

	myoepithelial marker cytokeratin 14 were expressed at higher levels in secondary compared to primary mammospheres (above), suggesting a shift in cell types and a possible enrichment for stem cells with the second round of culture. We also looked at the potential effect of radiation exposure to the donor mice on expression of these genes in primary and secondary mammospheres (below). All the genes examined were up-regulated in primary mammospheres from irradiated mice, but in the secondary mammospheres expression levels were beginning to return to the levels in unexposed controls. This may be a function of the additional ten days between irradiation of the animals and the measurement of gene expression in the secondary mammospheres, reflecting a gradual reduction in the radiation effect as time progresses. It may be that gene expression is not the most appropriate method for assessing relative stem cell enrichment, however. Further studies are underway to look at protein expression and immunofluorescent staining as potentially more informative endpoints. Animals have also been irradiated with SPE1972 Protons, 1 GeV/n 56Fe, and 350 MeV/n 28Si. Mammospheres are being cultured now, and will be injected in the next few weeks. Dysplastic lesions from these groups will be available for analysis around the end of the current funding period in January/February of 2012.
Bibliography Type:	Description: (Last Updated: 09/17/2020)
Articles in Peer-reviewed Journals	Meador JA, Ghandhi SA, Amundson SA. "p53-independent downregulation of histone gene expression in human cell lines by high- and low-LET radiation." Radiat Res. 2011 Jun;175(6):689-99. Epub 2011 Apr 26. PubMed PMID: 21520998 , Jun-2011